

**MATH 2551-K MIDTERM 1**  
**VERSION A**  
**FALL 2023**  
**COVERS SECTIONS 12.1-12.6, 13.1-13.4, 14.1**

**Full name:** \_\_\_\_\_

**GT ID:**\_\_\_\_\_

**Honor code statement:** I will abide strictly by the Georgia Tech honor code at all times. I will not use a calculator. I will not reference any website, application, or other CAS-enabled service. I will not consult with my notes or anyone during this exam. I will not provide aid to anyone else during this exam.

(     ) I attest to my integrity.

**Read all instructions carefully** before beginning.

- Print your name and GT ID neatly above.
- You have 75 minutes to take the exam.
- You may not use aids of any kind.
- Please show your work.
- Good luck! Write yourself a message of encouragement on the front page!



For problems 1-2 choose whether each statement is true or false. If the statement is *always* true, pick true. If the statement is *ever* false, pick false. Be sure to neatly fill in the bubble corresponding to your answer choice.

1. Any plane has only two distinct normal vectors.

☐ TRUE  
☒ FALSE

2. If  $\mathbf{u}$  and  $\mathbf{v}$  are vectors in  $\mathbb{R}^3$ , then  $\mathbf{u} \times \mathbf{v} + \mathbf{v} \times \mathbf{u} = \mathbf{0}$ .

☒ TRUE  
☐ FALSE

3. Order the following curves in increasing order of curvature.

- A) A helix of curvature 2  
B) A line segment of length 10  
C) A circle of radius 20

**Solution:**  $B < C < A$

4. Which of the following vectors could be the principal unit normal vector at time  $t = 2$  to a curve whose tangent line at  $t = 2$  is given by

$$\ell(p) = \langle 1, 0, -1 \rangle + p \langle 1, 2, 1 \rangle.$$

You must justify your answer to receive full credit.

- A)  $\langle 1, 1, 1 \rangle$   
B)  $\left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$   
C)  $\langle -2, 1, 0 \rangle$   
D)  $\left\langle \frac{-2}{\sqrt{5}}, \frac{1}{\sqrt{5}}, 0 \right\rangle$

**Solution:** The correct answer is D). Only B) and D) are unit vectors, and only D) is normal to the curve, since the dot product between it and the tangent vector  $\langle 1, 2, 1 \rangle$  at  $t = 2$  is 0, while the dot product with B is non-zero.

5. Tobert, a hard-working documentarian, is hiking through the African savannah for two days and his path is given by the curve  $\mathbf{r}(t) = \langle t^2, \frac{1}{3}t^3, 2 \rangle$  for  $0 \leq t < 48$ , where  $t$  is measured in hours and  $\mathbf{r}(t)$  in meters.

(a) What is Tobert's position two hours into his hike?

**Solution:**  $\mathbf{r}(2) = \langle 4, 8/3, 2 \rangle$

- (b) Find a function  $s(t)$  that gives the total distance Tobert has hiked for any time  $t$  since he started at  $t = 0$ .

**Solution:**

$$\begin{aligned} s(t) &= \int_0^t |\mathbf{r}'(\tau)| d\tau \\ &= \int_0^t |\langle 2\tau, \tau^2, 0 \rangle| d\tau \\ &= \int_0^t \sqrt{4\tau^2 + \tau^4 + 0} d\tau \\ &= \int_0^t \tau \sqrt{4 + \tau^2} d\tau \end{aligned}$$

Let  $u = 4 + \tau^2$ ,  $du = 2\tau d\tau$ . Then we have

$$\begin{aligned} s(t) &= \int_4^{4+t^2} \frac{1}{2} \sqrt{u} du \\ &= \frac{1}{3} u^{3/2} \Big|_4^{4+t^2} \\ &= \frac{(4 + t^2)^{3/2} - 8}{3} \end{aligned}$$

- (c) How far has Tobert hiked when he is at the point  $(4, 8/3, 2)$ ?

**Solution:** From a) and b), this is just  $s(2)$ .

$$s(2) = \frac{8^{3/2} - 8}{3} = \frac{8}{3}(2\sqrt{2} - 1)$$

6. In this problem, you will work with the differential equation

$$\mathbf{r}''(t) = 3\mathbf{i} + e^t\mathbf{j} + 49e^{7t}\mathbf{k}, \quad -\infty < t < \infty.$$

- (a) Find all vector-valued functions  $\mathbf{r}(t)$  with  $-\infty < t < \infty$  which are solutions to this equation.

*Hint: Your answer to this part should include some undetermined constant vectors.*

**Solution:** From the differential equation we get (via integration with respect to  $t$ ):

$$\mathbf{r}'(t) = 3t\mathbf{i} + e^t\mathbf{j} + 7e^{7t}\mathbf{k} + \mathbf{C}_1 \quad (1)$$

$$\mathbf{r}(t) = \frac{3}{2}t^2\mathbf{i} + e^t\mathbf{j} + e^{7t}\mathbf{k} + \mathbf{C}_1t + \mathbf{C}_2, \quad (2)$$

where  $\mathbf{C}_1$  and  $\mathbf{C}_2$  are constant vectors.

- (b) Find all vector-valued functions  $\mathbf{r}(t)$  with  $-\infty < t < \infty$  which are solutions to this equation and also have the property that  $\mathbf{r}'(0) = 3\mathbf{j} + 7\mathbf{k}$ .

**Solution:** Applying this condition to equation (1) gives

$$3\mathbf{j} + 7\mathbf{k} = \mathbf{r}'(0) = \mathbf{j} + 7\mathbf{k} + \mathbf{C}_1.$$

So we have  $\mathbf{C}_1 = 2\mathbf{j}$  and thus

$$\mathbf{r}(t) = \frac{3}{2}t^2\mathbf{i} + (e^t + 2t)\mathbf{j} + e^{7t}\mathbf{k} + \mathbf{C}_2. \quad (3)$$

- (c) Find all vector-valued functions  $\mathbf{r}(t)$  with  $-\infty < t < \infty$  which are solutions to this equation, have the property that  $\mathbf{r}'(0) = 3\mathbf{j} + 7\mathbf{k}$  and also have the property that  $\mathbf{r}(0) = \langle 1, 1, 5 \rangle$ .

**Solution:** Applying the new condition to equation (3) gives

$$\langle 1, 1, 5 \rangle = \mathbf{r}(0) = \langle 0, 1, 1 \rangle + \mathbf{C}_2.$$

So we have  $\mathbf{C}_2 = \langle 1, 0, 4 \rangle$  and thus

$$\mathbf{r}(t) = \left(\frac{3}{2}t^2 + 1\right)\mathbf{i} + (e^t + 2t)\mathbf{j} + (e^{7t} + 4)\mathbf{k}.$$

7. Let  $p_1$  be the plane defined by the equation  $2x - 3y + z = -10$  and let  $Q$  be the point  $(3, 1, 1)$ .

- (a) Find an equation for the plane  $p_2$  which contains the point  $Q$  and is parallel to  $p_1$ .

**Solution:** The normal vector to  $p_2$  needs to be parallel to the normal vector of  $p_1$ , so we can take the same vector. Since the plane must contain  $Q$ , we use that as the reference point:

$$2(x - 3) - 3(y - 1) + (z - 1) = 0 \quad \text{or} \quad 2x - 3y + z = 4.$$

- (b) Find an equation for the line  $\ell$  which passes through  $Q$  and is orthogonal to both planes.

**Solution:** Since  $\ell$  is orthogonal to both planes, its direction vector is parallel to the normal of both planes, so we again take the same vector and use  $Q$  as our reference point.

$$\ell(t) = \langle 2, -3, 1 \rangle t + \langle 3, 1, 1 \rangle.$$

- (c) Find the point  $R$  where the line  $\ell$  intersects the plane  $p_1$ .

**Solution:** To find  $R$ , we plug the line equation into the plane equation.

$$2(2t + 3) - 3(-3t + 1) + (t + 1) = -10$$

$$4t + 6 + 9t - 3 + t + 1 = -10$$

$$14t = -14$$

$$t = -1$$

So  $R$  is  $\ell(-1) = \langle 1, 4, 0 \rangle$ .

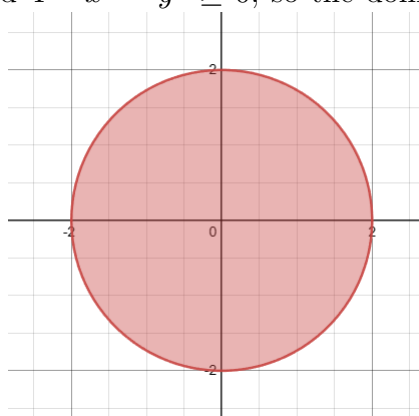
- (d) Compute the distance between the two planes using your work in parts (a)-(c) above.

**Solution:** The distance between the two planes is the length of any line segment orthogonal to both planes. Thus the segment of  $\ell$  between  $Q$  and  $R$  gives the distance:

$$|\vec{QR}| = \sqrt{(3-1)^2 + (1-4)^2 + (1-0)^2} = \sqrt{4+9+1} = \sqrt{14}.$$

8. (a) Find and sketch the domain of the function  $f(x, y) = \sqrt{4 - x^2 - y^2}$ . Show your work and label your axes clearly.

**Solution:** We need  $4 - x^2 - y^2 \geq 0$ , so the domain is all  $(x, y)$  inside or on the



circle  $x^2 + y^2 = 4$ .

- (b) Match the contour plots and graphs below with the given functions of two variables.

Function

Graph

Contour Plot

$$f(x, y) = \cos(\sqrt{x^2 + y^2})$$

\_\_\_\_\_

\_\_\_\_\_

$$g(x, y) = x^2 - y^2$$

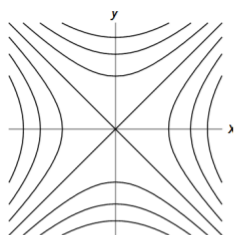
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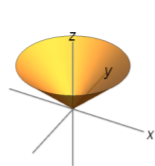
$$h(x, y) = x^2 + 2y^2$$

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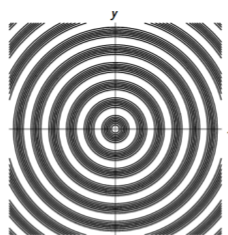
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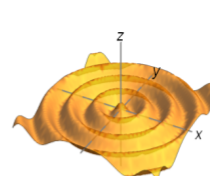
A



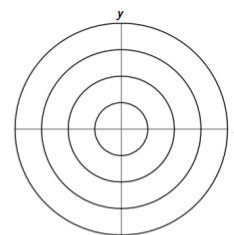
B



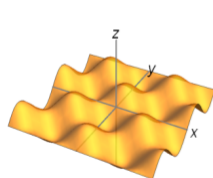
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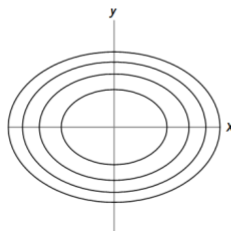
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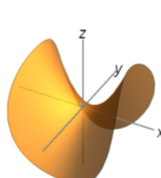
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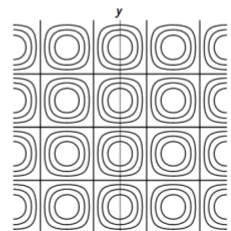
F



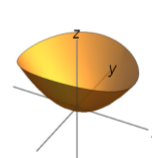
G



H



I



J

**Solution:**  $f$ : Graph D, Contour Plot C.

$g$ : Graph H, Contour Plot A.

$h$ : Graph J, Contour Plot G.

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**FORMULA SHEET**

- $\langle u_1, u_2, u_3 \rangle \cdot \langle v_1, v_2, v_3 \rangle = u_1 v_1 + u_2 v_2 + u_3 v_3$

- $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| |\mathbf{v}| \cos(\theta)$

- $\langle u_1, u_2, u_3 \rangle \times \langle v_1, v_2, v_3 \rangle = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$

- $|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}| |\mathbf{v}| \sin(\theta)$

- $L = \int_a^b |\mathbf{r}'(t)| \, dt$

- $s(t) = \int_{t_0}^t |\mathbf{r}'(T)| \, dT$

- $\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{d\mathbf{r}}{ds}$

- $\kappa = \left| \frac{d\mathbf{T}}{ds} \right| = \frac{1}{|\mathbf{v}|} \left| \frac{d\mathbf{T}}{dt} \right| = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}$

- $\mathbf{N} = \frac{1}{\kappa} \frac{d\mathbf{T}}{ds} = \frac{d\mathbf{T}/dt}{|d\mathbf{T}/dt|}$

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